Sky2Root

Test-drive your next crop rotation on 40 years of NASA data, before you plant.

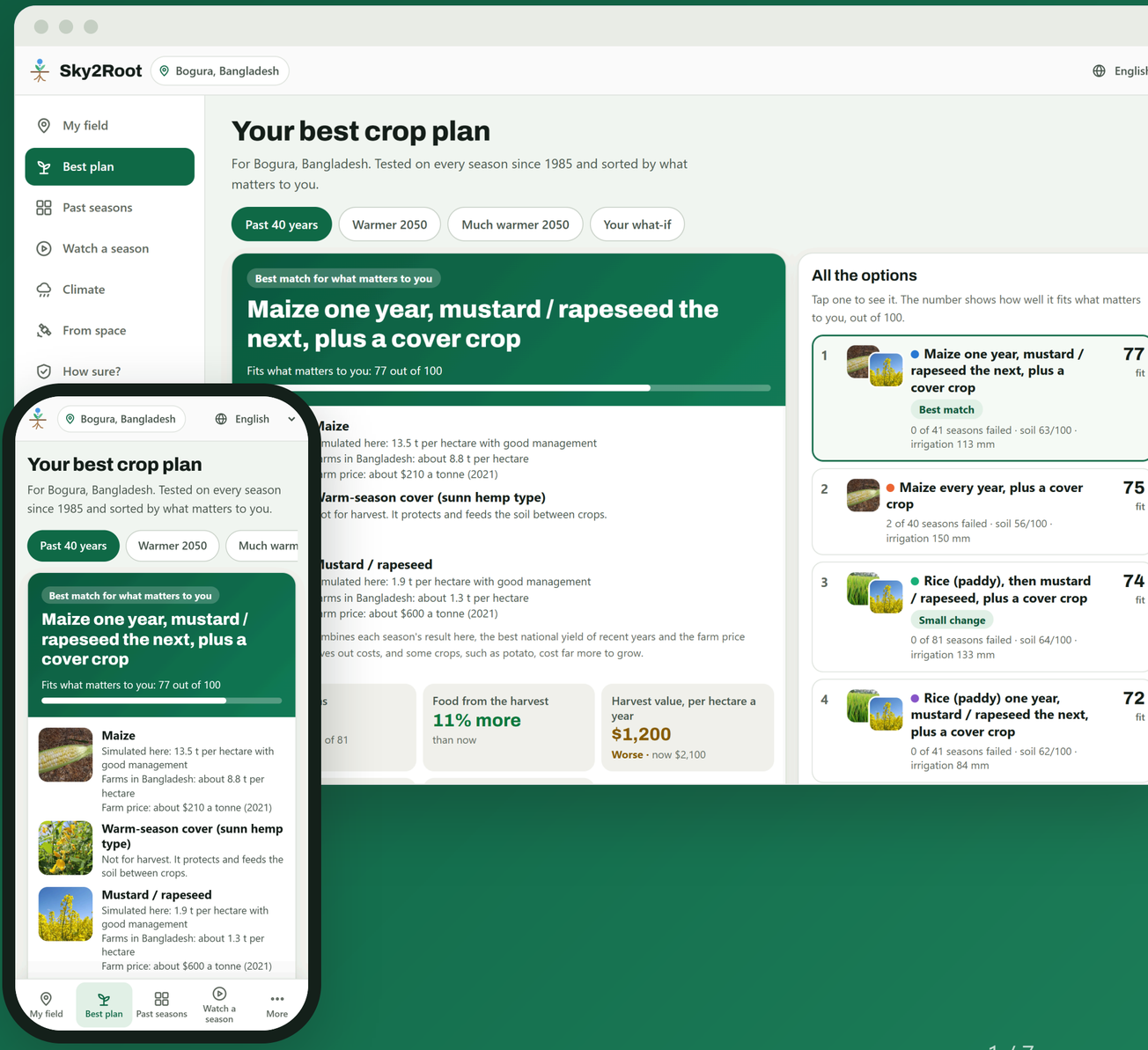
Team Groot · NASA Space Apps Challenge 2026

Challenge:

Field Shift: Adapting Farms with NASA Data

Try it: spaul571.github.io/sky2root

Code: github.com/spaul571/sky2root



Every screen in these slides is the working app.

THE PROBLEM

A farmer gets about 40 seasons in a lifetime. The climate is changing the rules.

Rain shifts

Late starts, long dry spells and floods break a calendar that worked for decades.

Heat at flowering

A few hot days at the wrong moment can empty a harvest.

Tired soil

Months of bare ground after harvest wash away in heavy rain.



A real field in Bogura, Bangladesh, month by month from space (NASA HLS: Landsat and Sentinel-2, through NASA GIBS). Green is growing crop; pale brown is bare soil.

NASA measures rain, heat, soil moisture and greenness for every field on Earth, but as satellite files, not answers. **Sky2Root turns them into a crop plan, for farmers and the advisers who serve them.**

HOW IT WORKS

Three questions. One answer.

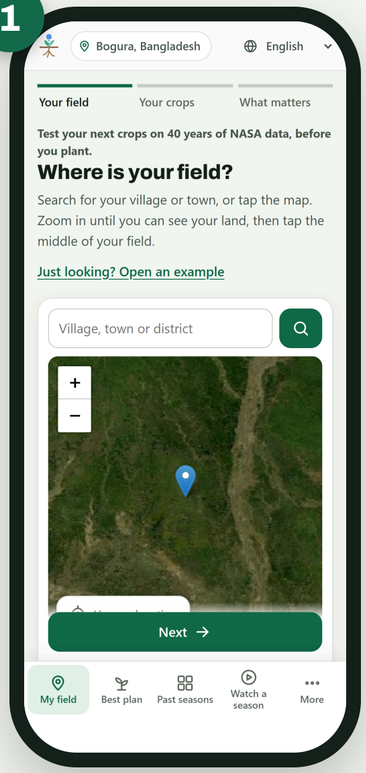
About 10 seconds for a new field

Free, no login

Works offline once opened

7 languages, incl. Bangla, Hindi, Kiswahili

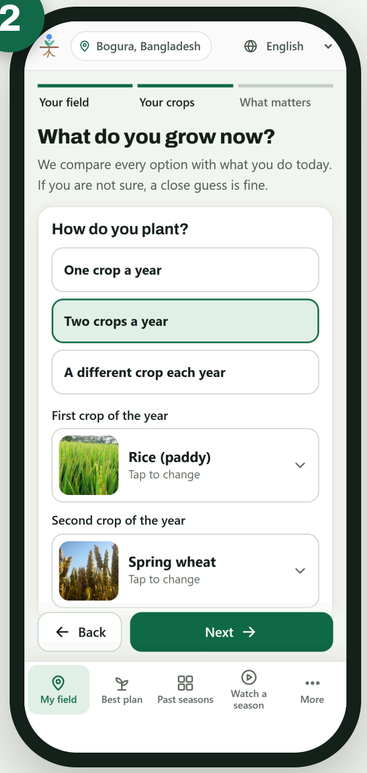
1



Where is your field?

Search, or tap the map. Any field on Earth.

2



What do you grow now?

Crops by photo; water and floods.

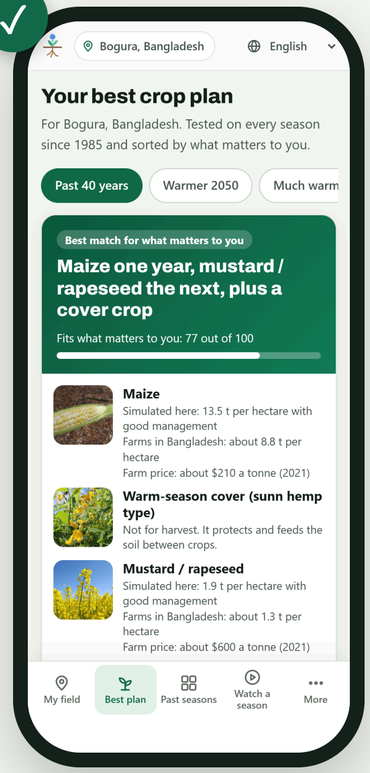
3



What matters most?

A sure harvest, money, water, soil, feed.

✓



Your best crop plan

Compared with what you grow now.

FROM THE SKY TO THE ROOT

Every season since 1985, replayed day by day

Sky NASA data for any point

NASA POWER daily heat, humidity, wind, sun since 1984

GPM IMERG daily rain since 1998

NEX-GDDP-CMIP6 2041-2060 from 5 climate models, 2 scenarios

SMAP L4 root-zone soil moisture, an independent check

HLS (Landsat, Sentinel-2), MODIS greenness at 30 m, bare-soil days, burned land, water use check

GRACE/GRACE-FO, SRTM water-storage trend, slope for erosion

→

Simulate each rotation

FAO-56 water balance rain fills the soil, the crop drinks it, day by day

Crop calendars sowing windows, heat at flowering, waterlogging, frost

Yield response FAO-33 and AquaCrop stress curves

Soil health living-root days, legumes, crop variety, erosion

Hotter 2050 the same seasons, shifted by each model

→

Root ranked for you

Your priorities a sure harvest, food, money, water, soil, feed

At least three plans the best match, a small change, and what you grow now

For each why it came up, the evidence, benefits, what is unknown, what to check locally

El Niño now how each plan did in past El Niño and La Niña years

Plain words in 7 languages, with every number traceable to its source

Different grids, one field: weather about 50 km, rain 10 km, soil 250 m, greenness 30 m, joined into one daily record for the point you mark.

Also: ISRIC SoilGrids 2.0 (soil), FAOSTAT (crops, yields and prices by country), NOAA Oceanic Niño Index, JRC Global Surface Water. Methods: docs/METHODS.md.

A REAL EXAMPLE: RICE LAND IN BOGURA, BANGLADESH

One small change: no failed seasons, more than twice the soil score

	WHAT THEY GROW NOW **Rice, then wheat**	SMALL CHANGE **Rice, then mustard, plus a cover crop**	BEST MATCH **Maize and mustard in turn, plus a cover crop**
Failed seasons, 1985-2025	**3 of 81**	**0 of 81**	**0 of 41**
Failed seasons in a warmer 2050 (SSP2-4.5; range across 5 models)	**11 of 81** 5-17	**0 of 81** 0-2	**0 of 41** 0-1
Soil health score	**27 / 100**	**64 / 100**	**63 / 100**
Irrigation a year	**144 mm**	**133 mm**	**113 mm**
Harvest value a hectare, before costs	**about $2,100**	**about $2,100**	**about $1,200** drops rice

Each plan comes with: Why it came up · The evidence · Possible benefits · What we do not know · What to check locally

Simulated on NASA POWER and GPM IMERG for this field, with default priorities. Yields assume good management, so read them as a comparison. Early temperatures drift at this site; the app flags it, and failure counts are a guide (docs/VALIDATION.md, check 9). A local expert should review any change.

CAN YOU TRUST IT?

Checked against data it never used

0.51

mean correlation of simulated soil water with NASA SMAP at 7 fields on 4 continents

12 of 15

known drought years flagged as poor or failed, including Iowa 1988 and 2012

4 of 4

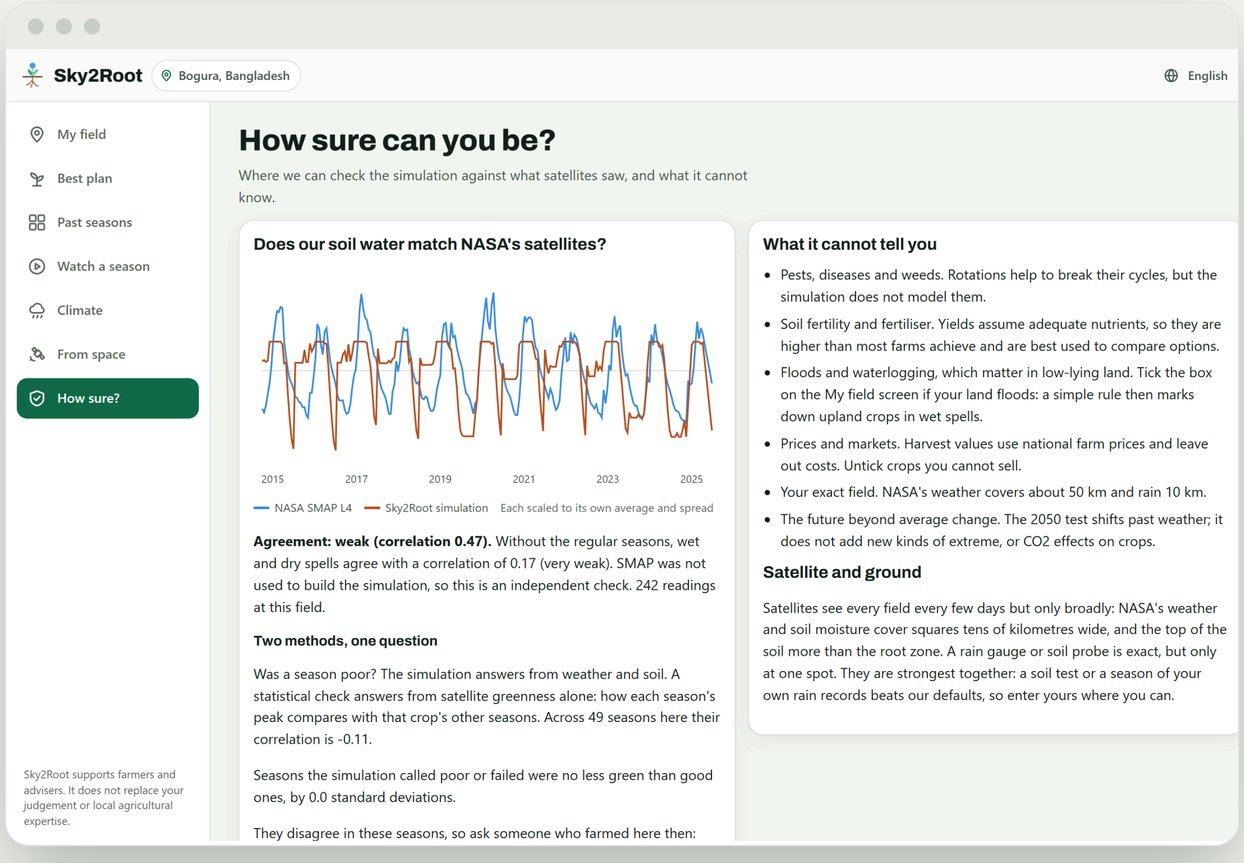
regions where the rain record shows El Niño's known effect

0

parameters fitted to these checks

Where it falls short is written down too: weak agreement with SMAP at flooded rice land, a drifting early temperature record, national yields in big countries. All in docs/VALIDATION.md.

It supports farmers and advisers. It does not replace their judgement.



TEAM GROOT

Next: into farmers' hands, with local experts

Shrikanta Paul	Team Lead
MD. Nurol Amin	Earth Data Engineer
Kazi Meherunnesa Eva	Crop and Climate Modeller
Fardin Ahmed Alvi	Frontend and UX Designer
Animesh Dey	Hardware and IoT Engineer

Try it	spaul571.github.io/sky2root
Code	github.com/spaul571/sky2root
Methods	docs/METHODS.md, docs/VALIDATION.md

What comes next

- Test the Bogura results with farmers and extension officers
- Native speakers to review the six other languages
- Field-level crop maps and a temperature record free of drift
- ECOSTRESS heat and water stress where it covers farmland
- Low-cost soil probes to check SMAP on the ground

Built with help from an SI coding assistant (Claude Code). The app itself uses no generative SI: its advice is worked out from the simulation by fixed rules.